OPTIMIZING PRICING STRATEGY WITH ROYALTY CONSTRAINTS

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Abstract. In this paper, we present a pricing optimization framework for determining minimum set prices across multiple publication formats under royalty and affordability constraints. Motivated by the not-for-profit mission of the MoKa Reads Collective—"spend the least to support the most"—our goal is to align publication pricing with fair compensation for the author to support themselves while maintaining accessibility for readers. We begin by analyzing price and page count data for programming books using the Google Books API and establish price boundaries for different formats. We then formulate a nonlinear optimization problem where the objective is to maximize expected royalties while penalizing excessive pricing. Constraints enforce pricing monotonicity and reward platforms with higher royalty rates. Using methods such as Sequential Least Squares Programming (SLSQP) and Trust-Region Constrained optimization, we demonstrate how our model provides a rational and principled method for price-setting across digital and print platforms. Our results show consistency between solvers and allow transparent, data-driven pricing decision for our publications.

In this paper we will be exploring a nonlinear optimization problem to determine the best prices for a publication given various distribution formats and producers. The method we use to determine the best price is by using a royalty constraint with the main premise being able to "spend the least to support the most", this slogan is directly tied to the collective's interest as a not-for-profit organization to sell publications not for pure profit, but in a affordable, accessible way that enables readers and our interest to being able to support ourselves aligned.

We will firstly discuss about the market, having a look at different books, their prices, page counts and trying to build a relationship with it, thereafter, we can start looking at our problem formulation, ways to approach it, and lastly we will look at results for determining the optimal prices for publications in the MoKa Reads Collective or a minimum set price.

1. Market Analysis and Price-per-Page Relationship

To begin, we analyze the current programming book market to explore the relationship between book price and page count. Although pricing can be influenced by factors such as author contracts, content demand, and publisher margin strategies, we restrict our focus to two primary variables: price and page count.

In this paper, all prices are assumed to be under the Canadian Dollar (CAD) unless explicitly stated not to be. Data was collected using the Google Books API, which provides convenient access to structured information such as book price and page count. This approach eliminates the need for web scraping and allows for efficient data retrieval under usage rate limits.

Given that the majority of our publications will be released in digital format, our pricing analysis focuses on ebooks. This is consistent with the format of our dataset, which exclusively contains ebooks. Print formats such as paperback and hardcover are considered secondary and are typically estimated by applying standard multipliers—1.5× for paperback over ebook, and an additional 1.5× for hardcover over paperback. For our purposes we will keep the price within the price bound we develop.

After cleaning the dataset to remove entries missing either price or page count, we obtained summary statistics shown in Table 1.

Statistic	Pages	Price ($CAD)
count	82.00	82.00
mean	553.34	79.22
std	288.75	66.19
min	91.00	8.99
25%	355.50	43.02
50%	481.50	55.99
75%	668.75	96.40
max	1488.00	314.94

TABLE 1. Descriptive statistics of page count and price for programming books retrieved from Google Books API

While the table provides a statistical overview, further insight is gained through visualization. Figure 1 shows a scatter plot of price versus page count distribution. A notable concentration of titles lies within the 200–400 page range, which we expect our publications in the collective to target.

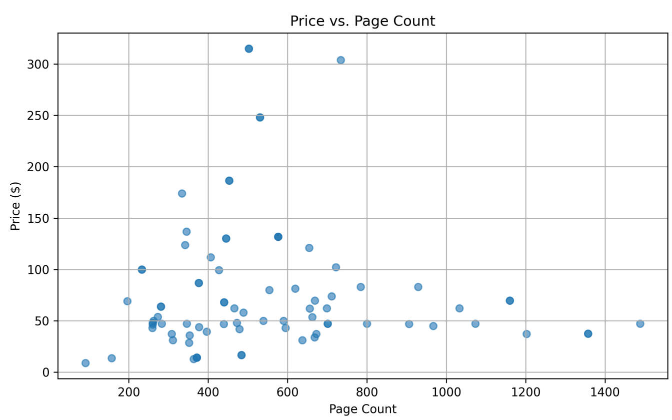

FIGURE 1. Scatter Plot of Price vs. Page Count

As illustrated in Figure 1, books within the 200–400 page range cluster around a price point of approximately \$50. This suggests that \$50 may serve as an upper bound for our formats, while pricing could reasonably target a midpoint range of \$25–30 to improve affordability.

2. The Optimization Problem

In the context of multi-platform publishing, determining optimal pricing across these various platforms and the formats they support (eg. Ebook, Paperback, Hardcover) requires a balance between the royalties earned per unit, format differentiation such as how ebooks won't have a print cost compared to paperback or hardcover which effects the author's earnings, and the affordability of the publication to the consumer or reader.

Let there be n publishing formats/platforms where each associated with a unit price $p_i \in [p_{\min}, p_{\max}]$, and a corresponding royalty rate $r_i \in (0,1)$ that determines the revenue earned per unit sold: $r_i p_i$. To ensure consistency and flexibility, we express each price as a normalized variable $x_i \in (0,1)$, such that:

$$p_i = p_{\min} + x_i(p_{\max} - p_{\min}) \tag{1}$$

This formulation allows the optimization to occur over the unit interval, while still mapping to actual price values within a global price bound.

The primary objective of our problem is to maximize the total expected royalty across all formats/platforms, while simultaneously penalizing excessive pricing to maintain affordability and market competitiveness, thus allowing consumers to be rewarded by choosing to buy the publication from platforms which offer higher royalty rates.The proposed objective function is a convex combination of maximizing total royalty and penalizing the total squared price:

$$\max_{\mathrm{x} \in [0,1]^{\mathrm{n}}} \quad \left[\sum_{i=1}^{n} r_i p_i - \lambda_p \left(\sum_{i=1}^{n} p_i \right)^2 \right] \tag{2}$$

Equivalently, this can be posed as a minimization problem for numerical optimization:

$$\min_{\mathrm{x} \in [0,1]^{\mathrm{n}}} \quad \frac{-\sum_{i=1}^{n} r_i p_i + \lambda_p \left(\sum_{i=1}^{n} p_i \right)^2}{n} \tag{3}$$

Here, the parameter $\lambda_p \geq 0$ governs the trade-off between maximizing roayalties and discouraging price inflation. A higher λ_p places greater emphasis on reducing the overall price, thereby improving affordability.

Constraints

To ensure logical structure and consistency in pricing and royalty assignment, we impose two key sets of contraints onto our problem:

1. **Monotocity in Price Levels:** To prevent illogical pricing where a format perceived to be of lower value is priced above a higher-tier format, we require strictly increasing normalized price levels:
$$x_{i+1} - x_i \geq \delta \quad \forall i = 1, ..., n-1 \tag{4}$$
2. **Monotocity in Royalty-per-Unit:** To incentivize distribution through high-royalty platforms and maintain economic alignment, we require that the effective royalty earned per unit decreases across formats of decreasing royalty rate:
$$r_i p_i \geq r_{i+1} p_{i+1} + \varepsilon \quad \forall i = 1, ..., n-1 \tag{5}$$

Initialization Strategy

To help acheive convergence faster and add bias the optimization towards realistic solutions, we propose a smart initialization based on the ordinal rank of the royalty rate. Let $R(r_i)$ denote the rank of format i's royalty rate among all formats, with rank 0 being the highest. Let U denote the number of unique royalty tiers, then the initial guess for each normalized price is given by:
$$x_i^{(0)} = \frac{R(r_i)}{U-1} \tag{6}$$
This initialization method places formats with higher royalty rates closer to the lower bound of the price range, thus having our model bias towards our affordability goal.

3. Applying the Model to Determine Minimum Set Prices

To solve the royalty-constrained pricing optimization problem, we employ gradient-based constrained optimization methods such as Sequential Least Squares Programming (*SLSQP*) and the Trust-Region Constrained (*trust-constr*) algorithm. These methods are well-suited for problems that feature nonlinear objective functions and nonlinear inequality constraints, as is the case here where we balance royalty maximization against pricing regularization while enforcing inter-format royalty and price orderings. SLSQP is particularly attractive due to its efficient handling of both equality and inequality constraints, along with variable bounds, in a relatively low-dimensional space. Alternatively, the trust-constr method provides a more robust framework by incorporating trust-region steps and allowing for Jacobian and Hessian approximations, which improves stability in the presence of tightly coupled constraints. Both methods leverage the smoothness and structure of the problem to converge efficiently to locally optimal solutions that satisfy all pricing and royalty conditions. This makes them strong candidates for reliably determining minimum viable prices across publishing formats while maintaining economic consistency.

Let us consider the following platforms that are used for self-publishing by the MoKa Reads Collective: MoKa Reads Shop, Kindle Direct Publishing (KDP), Leanpub, Kobo, Google Books and Barnes & Noble (B&N). These plaforms have the following royalty rates, it is assumed the format is Ebook, unless explicitly stated otherwise:

Platform	Royalty
MoKa Reads Shop	87%
Leanpub	80% [1]
Kobo, Google Books, B&N	70%
KDP Paperback	60% [2]
B&N Print	55% [3]
KDP Ebook	35% [4]

Table 2. Platform Format Royalty Rates

It is important to note that the reason our royalty rate for KDP is 35% is due to our ebook prices being over the maximum of 9.99 that KDP requires to be eligible for its 70% royalty option. Another thing to remember is the royalty-per-unit for print formats (paperback, hardcover) is the earnings from the royalty, and does not include printing costs, so while for ebook, we can synonymously say the royalty is the profit margin, the same cannot be said for print formats, and is the reason we say the prices are the minimum set, while the optimal prices for ebooks will not change if the book is between 200-400 pages, the same guarantee cannot be made for print formats, and may be be adjusted due to print costs.

The following parameters are used in our optimization solver:

Parameter	Value
$(p_{\min}, p_{\max})$	$(8.99, 50)$
δ	0.05
ε	0.25
λ_p	0.05
r	$[0.87, 0.8, 0.7, 0.6, 0.55]$

Table 3. Parameters for SLSQP and TRCP solvers

With these parameters we have the following results for each corresponding format of the royalty rate of the platform:

Format	Price (SLSQP)	Royalty-per-Unit (SLSQP)	Price (TrustRegion)	Royalty-per-Unit (TrustRegion)
Format 1	27.01	23.49	27.01	23.5
Format 2	29.06	23.24	29.06	23.25
Format 3	31.11	21.77	31.11	21.77
Format 4	33.16	19.89	33.16	19.89
Format 5	35.21	19.36	35.21	19.36
	155.53	107.77	155.53	107.77

TABLE 4. Comparison of optimal prices and royalties obtained using SLSQP and Trust-Region Constrained methods for multi-format royalty-constrained pricing.

Appendix

A. Pseudo-Code.

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Inputs:
    n          ← number of publishing formats
    r[1..n]    ← royalty rates per format
    p_min      ← global minimum price
    p_max      ← global maximum price
    λ_p        ← price penalty weight
    δ          ← minimum price spacing (normalized)
    ε          ← minimum royalty-per-unit spacing

Initialize:
    For each format i = 1 to n:
        Assign normalized price variable x_i ∈ [0, 1]
        Set initial guess x_i^(0) ← rank(r_i) / (R - 1)

Define:
    p_i(x_i)     ← p_min + x_i * (p_max - p_min)
    objective(x) ← ( -∑ r_i * p_i(x_i) + λ_p * (∑ p_i(x_i))^2 ) / n

Subject to constraints:
  For i = 1 to n - 1:
    x_{i+1} - x_i ≥ δ                                [Price ordering]
    r_i * p_i(x_i) ≥ r_{i+1} * p_{i+1}(x_{i+1}) + ε  [Royalty ordering]
    For all i:
        0 ≤ x_i ≤ 1                                  [Normalized bounds]

Solve:
  Use a constrained optimization algorithm (e.g., SLSQP or Trust-Region
Constrained) to find x* = argmin objective(x) subject to constraints

Output:
    For each i:
        Price:   p_i = p_min + x_i * (p_max - p_min)
        Royalty: r_i * p_i
    Report:
        Total Price   = ∑ p_i
        Total Royalty = ∑ r_i * p_i
```

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